IGCSE 0607 Practice Questions — E3.4 — Combined Questions & Markscheme

Finding a quadratic function using given information

Total Questions: 3 — **Total Marks:** 13 Marks

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1 A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) The graph passes through the point $(5, p)$.
Work out the value of p.

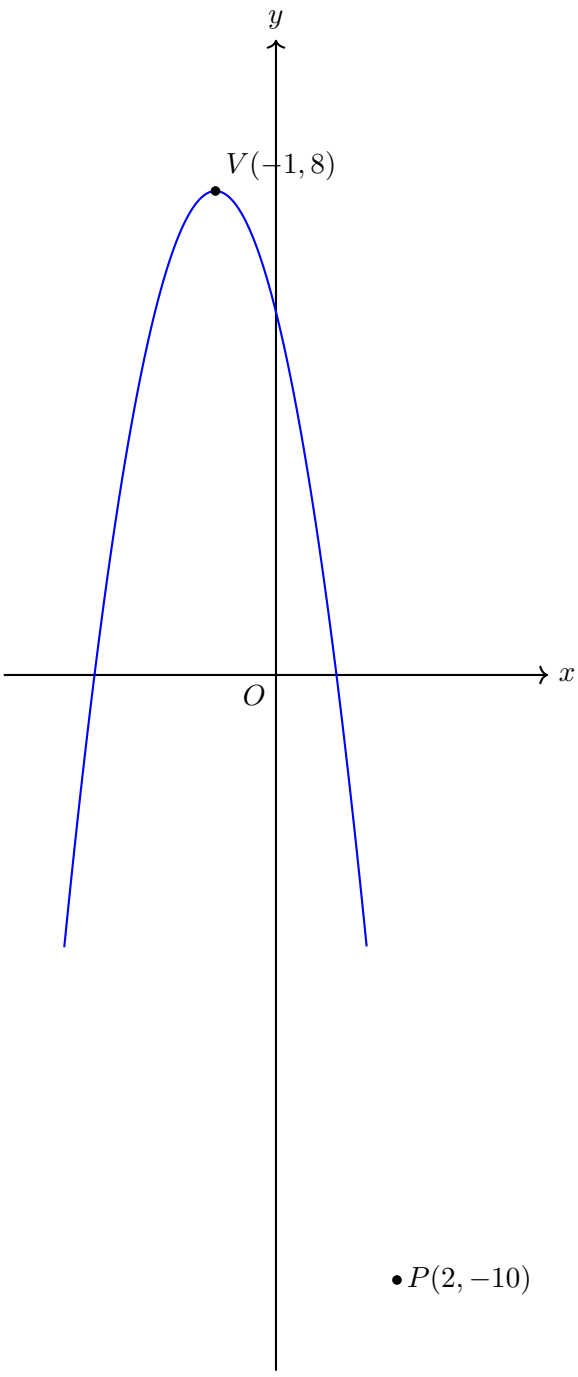
$p =$ ..... **[2]**

2 The graph of a quadratic function $y = f(x)$ intersects the x-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the y-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

3 The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.



Work out the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

1 **(a)** $h = 3$ **[B1]**

$k = -2$ **[B1]**

(b) $p = (5-3)^2 - 2$ **[M1]**

$p = 2$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 $f(x) = a(x+2)(x-5)$ **[M1]**

$a(0+2)(0-5) = -20$ or $-10a = -20$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 20$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 $y = a(x+1)^2 + 8$ **[M1]**

$-10 = a(2+1)^2 + 8$ **[M1]**

$9a = -18 \implies a = -2$ **[A1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$a = -2,\ b = -4,\ c = 6$ **[A1]**

[Back to Question] *[Total Marks: 5]*